

Comparing EEG and fMRI in investigating the Bayesian brain hypothesis: insights into predictive processing

The Bayesian brain hypothesis provides a powerful computational framework for understanding perception and cognition, suggesting that the brain functions as a probabilistic inference machine by continuously generating and updating internal models to interpret sensory input and optimise behaviour under uncertainty (Friston, 2012). Drawing on Bayesian statistics, this theory proposes that the brain integrates prior expectations with current sensory evidence to generate predictions about the environment and adjusts those predictions when they are violated. This process is formalised in predictive coding models, which posit a hierarchical neural architecture in which top-down predictions are compared with bottom-up sensory signals, and the resulting prediction errors are used to refine internal models over time (Rao & Ballard, 1999; Friston, 2005; Bastos et al., 2012). These mechanisms are not confined to perception alone but are thought to underlie a wide range of cognitive functions including attention, learning, and decision-making (Bottemanne, 2021).

Importantly, the Bayesian brain hypothesis makes explicit, testable predictions about neural processes, such as where and when predictions and prediction errors are represented in the brain. These questions necessitate neuroscientific methods that can capture both the temporal dynamics and spatial localisation of brain activity. Electroencephalography (EEG) and functional magnetic resonance imaging (fMRI) are two complementary tools in cognitive neuroscience that have been extensively used to examine predictive processing. EEG offers high temporal resolution, allowing researchers to track the rapid unfolding of prediction and error signals over milliseconds. In contrast, fMRI provides high spatial resolution, enabling the identification of brain regions involved in generating predictions or encoding mismatches. Together, these methods allow researchers to empirically test key components of the Bayesian framework such as hierarchical processing, precision weighting of prediction errors, and top-down modulation. This offers converging insights into how the brain implements probabilistic inference. This essay critically evaluates EEG and fMRI as tools for investigating the Bayesian brain hypothesis, comparing their respective strengths, limitations, and contributions to our understanding of predictive coding in the human brain.

fMRI is a non-invasive neuroimaging technique that indirectly measures brain activity by detecting changes in blood oxygenation. When neural populations become more active, they consume more oxygen, prompting localised increases in blood flow that can be spatially resolved using fMRI. This method offers high spatial resolution, enabling precise localisation of activity across cortical and subcortical regions. However, its temporal resolution is limited to the scale of seconds, making it less effective for tracking fast neural dynamics. In the context of the Bayesian brain hypothesis, fMRI provides critical insights into where in the brain predictive processes occur.

For instance, reduced blood-oxygen-level-dependent (BOLD) responses in the fusiform face area for expected versus unexpected stimuli (Summerfield & Egner, 2009) suggest that accurate predictions suppress bottom-up error signalling. Similarly, den Ouden et al. (2012) demonstrated that prediction error signals are modulated by prior expectations and encoded across sensory and frontal regions. These findings support predictive coding models by showing that neural responses are shaped by prior beliefs and that hierarchical, top-down modulation occurs across distributed brain networks.

EEG is a non-invasive technique that records electrical activity from the scalp, offering millisecond-level temporal resolution ideal for capturing the fast dynamics of perceptual and cognitive processes. While its spatial resolution is limited, EEG excels in identifying when neural events occur. This makes it particularly valuable for testing the Bayesian brain hypothesis, especially in detecting prediction error signals in real time. A key method involves analysing event-related potentials (ERPs), such as the mismatch negativity (MMN) which is elicited when an auditory stimulus violates predicted patterns. It is interpreted as a marker of bottom-up prediction error (Näätänen et al., 2007). In the visual domain, attenuated early ERP components (e.g., P1, N1) during expected stimuli also support predictive suppression effects. Time-frequency analyses further reveal that different neural oscillations track distinct aspects of predictive processing; alpha and beta bands are linked to the encoding and maintenance of prior expectations (Arnal & Giraud, 2012), while gamma activity is associated with feedforward transmission of prediction errors (Bastos et al., 2015). Together, EEG findings provide strong temporal evidence for predictive coding, complementing fMRI by elucidating the dynamic unfolding of Bayesian inference in the brain.

The contrasting resolution profiles of EEG and fMRI have important implications for how each method contributes to our understanding of predictive coding. EEG's superior temporal resolution makes it uniquely suited to track the unfolding of predictions and prediction errors in real time.

These findings offer strong support for the view that top-down expectations exert rapid influence on perception, as proposed by Bayesian models. In contrast, fMRI's strength lies in its ability to localise the neural substrates of these predictive processes. For example, studies have shown that expected stimulus elicit reduced BOLD responses in regions such as the fusiform face area and primary visual cortex, consistent with efficiency prediction reducing bottom-up error signals (e.g., Summerfield & Egner, 2009). Moreover, fMRI has revealed the involvement of higher-order regions like the prefrontal cortex in modulating activity in lower-level sensory areas, providing spatial evidence for hierarchical predictive coding. Thus, while EEG clarifies when predictions take place. The complementary strengths of both methods are therefore essential to fully capturing the spatiotemporal dynamics of Bayesian inference in the brain.

EEG and fMRI differ not only in the type of data they provide but also in the psychological inferences they afford. EEG's fine-grained temporal resolution allows researchers to infer the timing and sequence of cognitive processes involved in prediction. For instance, ERP studies using auditory or visual oddball paradigms have shown that unexpected stimuli elicit early prediction error signals, suggesting that the brain constantly generates and updates internal models to anticipate incoming input. This supports dynamic, trial-by-trial models of Bayesian inference, in which perceptual systems rapidly adjust based on violated expectations. Conversely, fMRI allows stronger inferences about the localisation and interaction of brain systems involved in predictive processing. Studies using fMRI have identified distinct cortical hierarchies engaged in encoding prior beliefs and computing prediction errors, allowing researchers to infer how different brain regions collaborate to implement top-down and bottom-up signalling. Furthermore, the spatial resolution of fMRI has enabled the mapping of predictive coding mechanisms across sensory and associative cortices, reinforcing the hierarchical structure central to the Bayesian brain framework. Ultimately, EEG offers insight into the temporal architecture of prediction, while fMRI provides evidence for the anatomical pathways supporting this function. Both methods thus enable complementary psychological inferences; EEG clarifies how and when predictions influence cognition, whereas fMRI elucidates where these processes are instantiated in the brain.

The combined findings from EEG and fMRI research have significantly shaped the theoretical development of the Bayesian brain hypothesis. EEG contributes crucial support for the idea that perception involves dynamic, temporally precise predictive mechanisms. The detection of early ERP components such as MMN, and their modulation by expectation, provides direct evidence for rapid, automatic error signalling which is a core tenet of predictive coding theories. This aligns with the notion that the brain continuously generates forward models to anticipate sensory input, updating them as discrepancies arise. Meanwhile, fMRI has clarified how the computations are implemented within a hierarchical cortical architecture. By revealing which brain regions encode priors and errors, and how higher-order areas modulate activity in sensory cortices, fMRI supports the hypothesis that predictive coding operates through bidirectional signalling across multiple levels of the brain. Importantly, both methods have helped refine the Bayesian model from a theoretical abstraction to a biologically plausible framework. EEG demonstrates the temporal feasibility of rapid prediction-error signalling, while fMRI shows the anatomical plausibility of a distributed prediction network. Together, they provide converging evidence that the brain is not a passive receiver of stimuli, but an active inference system guided by probabilistic expectations.

References

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